Assignment - 4

Data Story (Group)

MBAI 5400G: Visualization & Storytelling

Winter 2026

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Prepared By: Group 3

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1.Introduction: Our Research Background

Artificial intelligence is increasingly affecting the workplace. Some occupations, for example administrative works, data entry, customer service, and business support roles can be more exposed to AI than others.

Not only these, in recent times housing prices in Oshawa have increased significantly. This creates a serious concern for middle-aged workers. If their jobs become unstable or require retraining, they will struggle to keep up with rising housing costs.

This is why we as a group decided that we will be choosing this topic which actually shows impact of AI to Oshawa middle aged workers and its future impact on increasing housing prices. So we decided our research title as follows:

"Estimating AI Exposure in Oshawa Workforce and Its Impact on Housing Affordability for Middle-Aged Workers"

The main idea of this research is that some occupations in Oshawa can be more exposed to AI than others, and workers in those occupations will face employment uncertainty, wage pressure, or the need for retraining. If housing prices continue to rise while workers face these challenges, affordability will become more difficult.

Our analytical flow is:

Workforce structure in Oshawa
↓
AI exposure across occupations
↓
Estimated number of workers affected by AI
↓
Housing price growth in Oshawa
↓
Overall workforce AI risk and affordability pressure

2.Our Research Grounding Question:

When AI Changes the Job Market, Can Middle Aged Workers in Oshawa Still Afford a Home? So in our research we mainly tried to find about in what extent is the Oshawa workforce exposed to artificial intelligence. Our work has focused on occupations in Oshawa that are most vulnerable to AI-related disruption. We also have shown that in this AI impact timeline the housing prices are increasing. Thus we have tried to show that the AI exposure made job vulnerable and in upcoming years these impact to middle age workers home affordability. Based on these thinking, we made a research proposal and submitted it and then we started to work on our final outcome.

3.Methodology:

As a group we have decided to follow a process to give the actual outcome.

3.1 Initial Sketches

First we took our pen and paper and made some sketches about how we will approach. To co-ordinate among group members we used Google Drive and WhatsApp group for easy file sharing and daily communication.

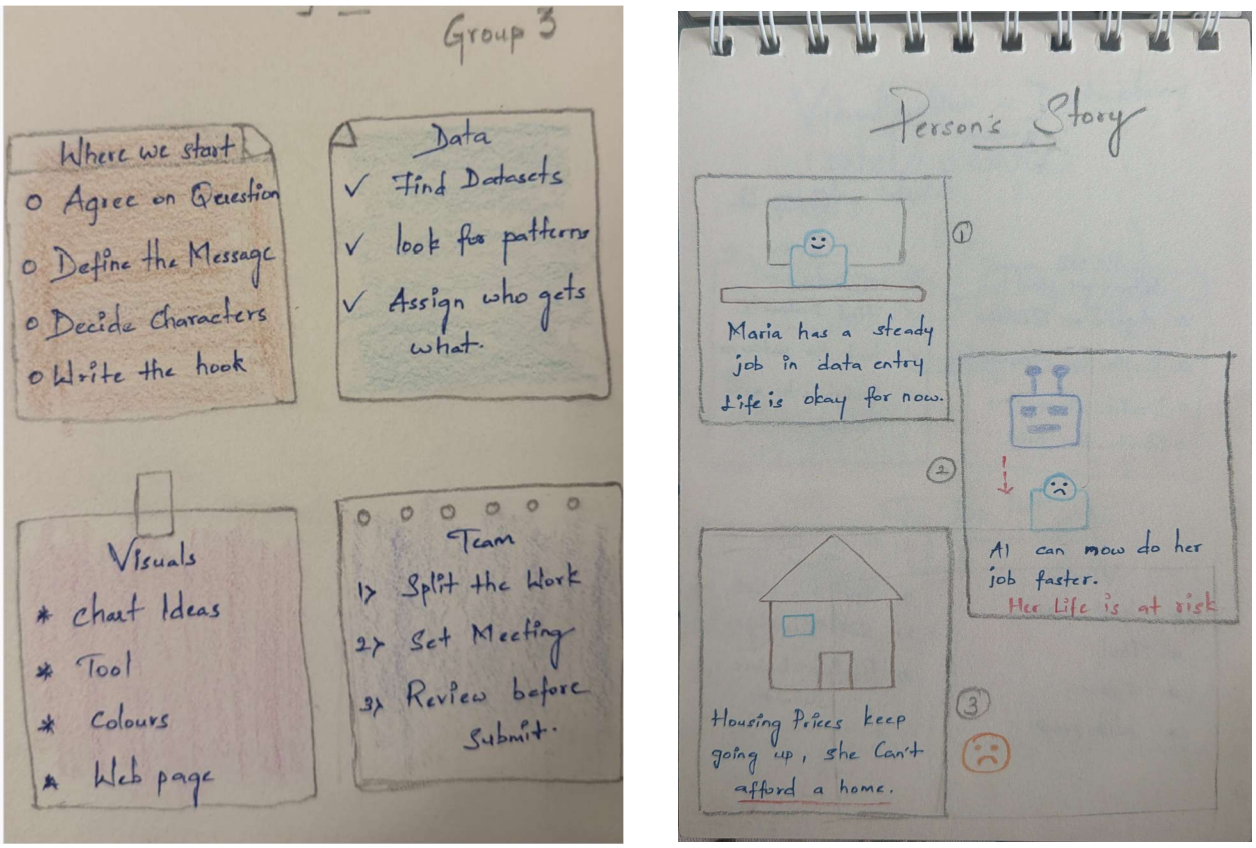

Figure 1.1: Our Initial Approach

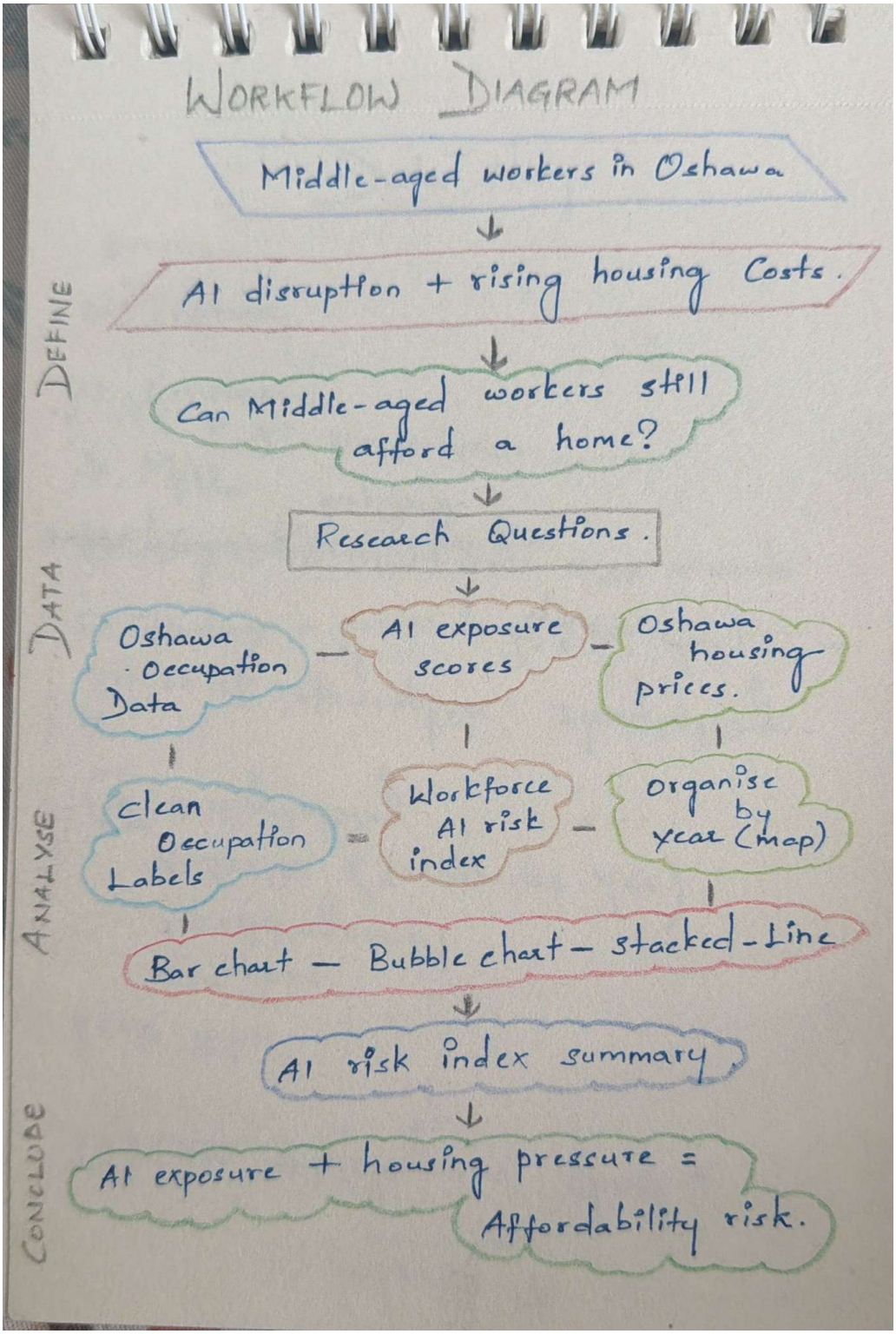

Figure 1.2: Initial Sketch of Workflow Diagram

After this draft sketches, we used physical sticky notes to make a workflow chart to follow. The physical sticky note diagram (Figure 2) shows the steps we followed. For example, after

finalizing the theme we started our final flow diagram with MIRO board (Figure 3). Then we started researching about actual characters who got affected in real life. Then we started working on possible data sources and downloading those data sets and studying those. As we decided our audience will be general people and the policy maker both, we started working on how to visualize the data and taking decision which chart or methods will be best. After that we started implementing our scrolly telling method.

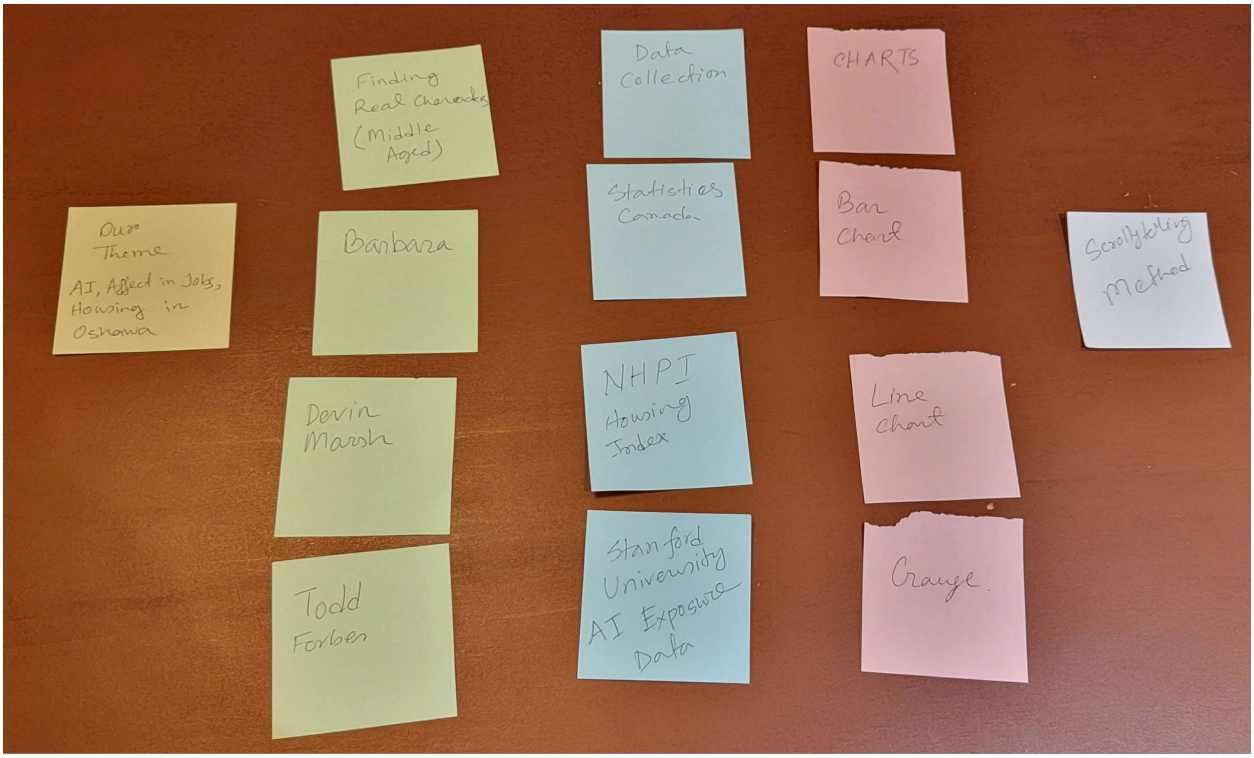

Figure 2: Physical Sticky Notes

<u>3.2 Miro Board:</u>

The Miro board on Figure 3 has been used as a structured, flow-based ideation tool to map out our project on AI, job disruption, and housing affordability in Oshawa for middle-aged workers. Following a left-to-right layout, the board has been organized into key stages: themes, characters, datasets, data analysis, visualizations, and final format. We have begun by defining our central theme, highlighting rising housing costs, automation risk, and increasing economic pressure on workers. From there, we have identified key personas—from a middle-aged worker like an auto parts supplier, a telecommunications receptionist, and a customer service agent—to represent individuals most affected by these changes. These characters have been connected to a central node, "Real People Affected in Oshawa," reinforcing the human-centered focus of our story. The board then has mapped how multiple data sources such as

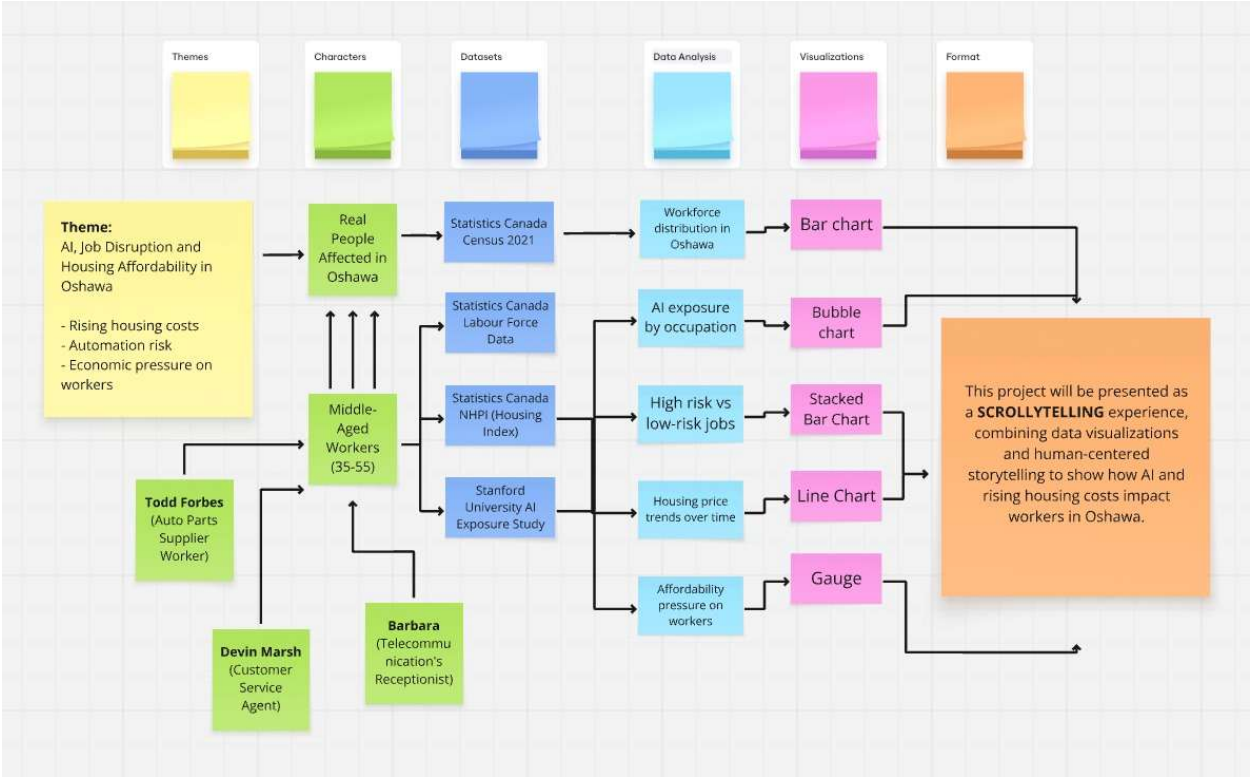

Figure 3: MIRO Board

Statistics Canada (Census 2021, Labour Force data, and NHPI) and the Stanford AI exposure study have fed into our analysis stage, where we have categorized insights including workforce distribution, AI exposure by occupation, job risk levels, housing price trends, and affordability pressures. Each of these insights has been directly linked to a specific visualization type, such as bar charts, bubble charts, stacked bar charts, line charts, and a gauge for key metrics. Finally, all elements have converged into a scrollytelling format, combining data visualization and narrative storytelling. This structured flow has ensured a clear connection between data, insights, and the final user experience.

3.3 Character

When we have been discussing about how we will collect our data, one of our group member mentioned that her neighbor lost her job due to AI. So our group member decided to talk with her neighbor. For personal reason we did not use her neighbors real name or picture.

3.3a Barbara:

When our group mate has spoken with Barbra, a neighbour who has lost her job to AI, the issue has stopped being an abstract idea or something we simply read about. Barbara expressed the uncertainty and vulnerability in her voice, along with a quiet strength and determination to keep moving forward. That moment has will stay with us. It has shown that what we often reduce to numbers and statistics have been lived experiences.

Alongside this personal conversation, we have also explored publicly shared stories reported in the media, including accounts featured by CBC and Canadian Affairs.

3.3b Todd Forbes

We have also learned about the experience of Todd Forbes, whose story has reflected the uncertainty facing manufacturing workers in Oshawa. A lifelong resident of the city, he has been among the roughly 2,000 workers affected by job losses linked to changes at the General Motors Oshawa plant. After working at TFT Global Inc., an auto-parts supplier connected to the plant, he has faced the prospect of unemployment as the midnight shift has been cut. With deep family roots in Oshawa and strong ties to the community, he has described the situation as gut-wrenching, yet he has begun considering relocating to Nova Scotia in hopes of finding stable employment. His experience has highlighted how industrial changes, job insecurity, and regional economic pressures have been shaping difficult life decisions for middle-aged workers and their families.

Figure 4: Actual Image of Todd Leaving Oshawa

3.3c Devin Marsh

Through this reporting, we have learned about the experience of Devin Marsh, whose story has illustrated the real impact of AI on customer service work. He has worked as a senior customer support and sales agent for Rogers through a third-party company since 2018, and when an AI

tool has been introduced in 2024, it has initially been presented as a way to make his job easier. Over time, however, the tool has become mandatory, and he has found himself training the very system that would eventually replace his role. Within months, large-scale layoffs have begun, with more than 1,000 employees having been let go, and he has lost his job in late September while working remotely from Oshawa. His experience has captured the emotional shock of displacement, describing it as feeling like the rug has been pulled out from under him, and it has highlighted how customer service roles have become especially vulnerable to AI-driven automation. In the news source Devin's picture was not shown but the work setup photo (Figure 5) was shown.

Figure 5: A work setup similar to Devin Marsh

3.4 Data Collection

We collected data from several trusted sources to understand Oshawa's workforce, AI risk, and housing trends. Workforce information came from Statistics Canada's 2021 Census (oshawa_census_2021_occupation_data.csv), which provides detailed counts of workers across the 10 major NOC 2021 job categories. This allowed us to see which types of jobs are most common in Oshawa and how many people are employed in each sector. To measure how AI could affect these jobs, we used data from Stanford HAI's AI Index Report 2025 (occ_level.csv on GitHub). This dataset includes AI exposure scores for 923 occupations from the U.S. O*NET database, combining algorithmic ratings and human expert judgments. We mapped these occupations to Canada's NOC system to estimate AI risk for Oshawa workers. For housing

trends, we used Statistics Canada's New Housing Price Index (NHPI, Table 18-10-0205-01), which tracks the cost of new homes over time. This helped us understand how rising housing prices overlap with changes in the labor market. By combining these sources, we can tell a complete story about how artificial intelligence and affordability are affecting workers in Oshawa.

3.5 Data Visualization / Charts:

We have designed our charts in Flourish to test how they look and to decide which format is easiest to understand. At first, we have tried a scatter plot, but it has been difficult to read, so we have switched to a heat map with a color gradient, which has made the patterns much clearer and more attention-grabbing. We have also created a bar chart to show the number of workers in each occupation category, a stacked bar chart to compare workers in AI-exposed versus non-exposed jobs, and a line chart to illustrate changes in housing prices over time. Creating these visualizations has not always been easy as errors have kept appearing in the notebook, which has been frustrating and stressful, especially with limited time. But working through these challenges has helped us feel more connected to the data and the story it tells. Watching the charts come together, the message has become clear that many people work in jobs that could be affected by AI while housing costs continue to rise. The visualizations have made the data feel real and human, not just numbers on a page.

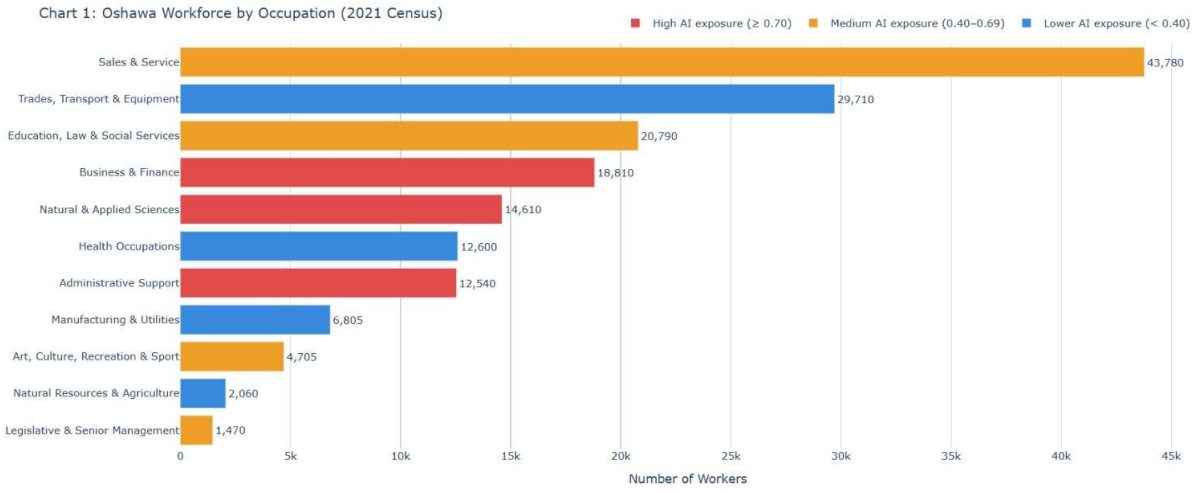

Figure 6.1: Oshawa Workforce Chart

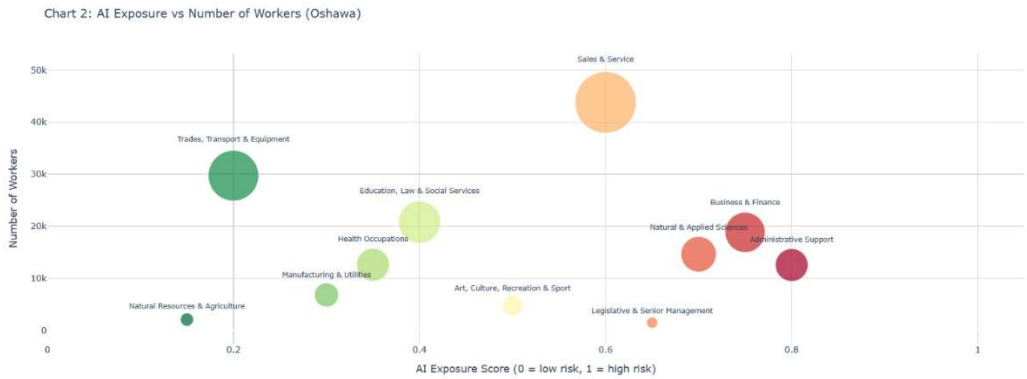

Figure 6.2: AI Exposure Chart

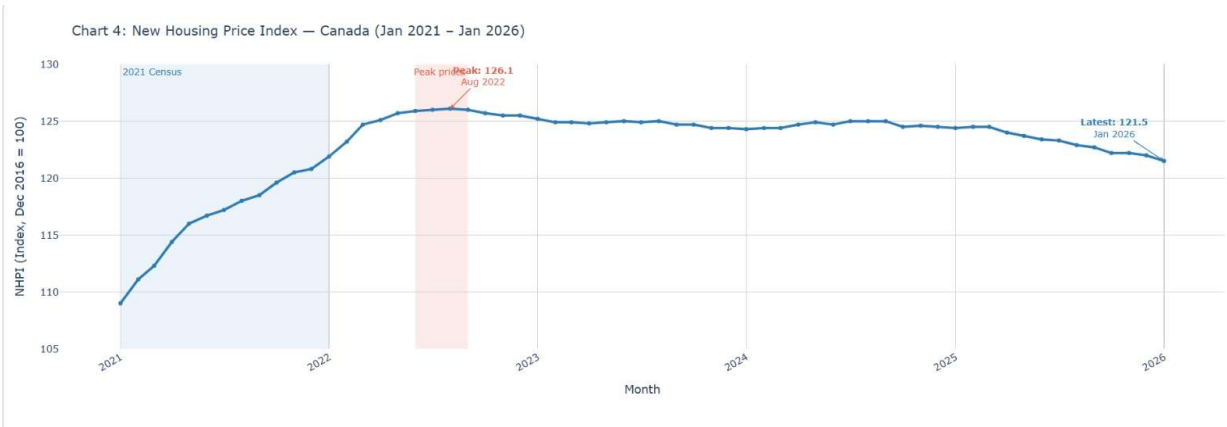

Figure 6.3: Housing Price Index

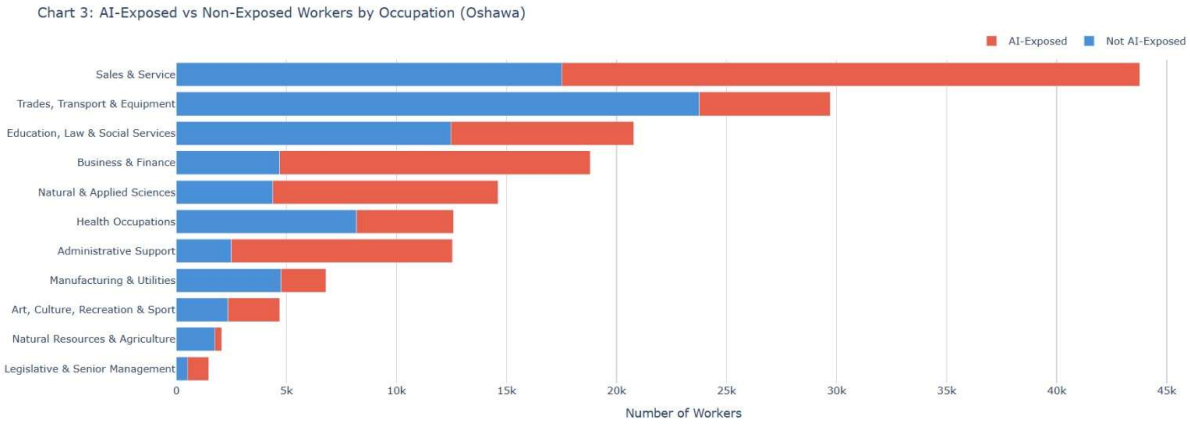

Figure 6.4: Exposed Vs Non Exposed Chart

After making these charts, we as a group decided that the comparison chart we are gonna make it as radial bar. We also decided that the scatterplot was hard to understand so we selected color gradient heat map to make it easier to understand for our audience.

3.6 Data Calculation

We have estimated how many workers might be affected by AI. We have linked AI scores from O*NET jobs (stanford_hai_923_occupations_ai_scores.csv) to Oshawa's job categories (oshawa_census_2021_occupation_data.csv). For each job category, we have calculated a weighted average AI exposure using the formula:

AI_Exposure_NOC = Σ(AI_Score × Worker_Count) / Σ(Worker_Count)

Then we have calculated Workers_Affected = Total_Workers × AI_Exposure_Percentage. This shows that about 49,000 workers, or 29% of Oshawa's workforce, may face AI-related changes. We have also calculated an overall AI Exposure Index for all jobs. Housing price data has been taken from a CSV file and used directly to show changes over time.

3.7 Implementation: Scrollytelling

After all these we started our implementation. We chose Scrolly-Telling method. Our first motto was we will show the reall affected peoples story and then will show the data charts. Audience will scroll through, they will get emotionally connected and then they will see the real data to validate our story. Below we are attaching how we decided to divide our page to tell the story.

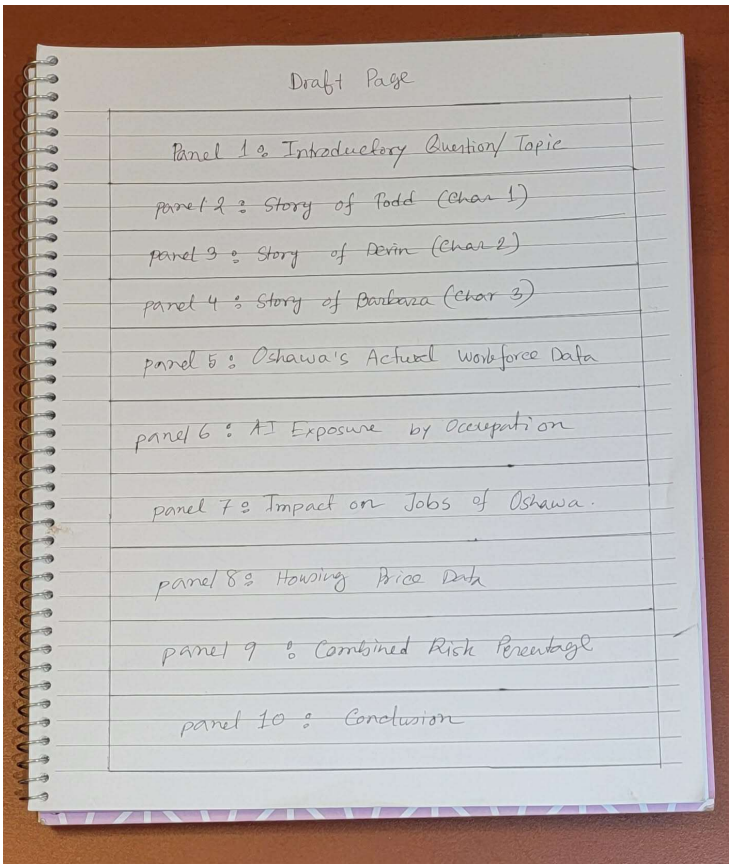

Figure 7: Our Page Layout Draft

After finalizing our design, we implemented our web page which will tell the story while scrolling through it. We used web templates to make or page. We also tried to put transitions to make it more modern. All the charts are interactive, audience can click/hover to see data at a specific point.

We are attaching screenshot of our final page below:

Figure 8.1: Final Output

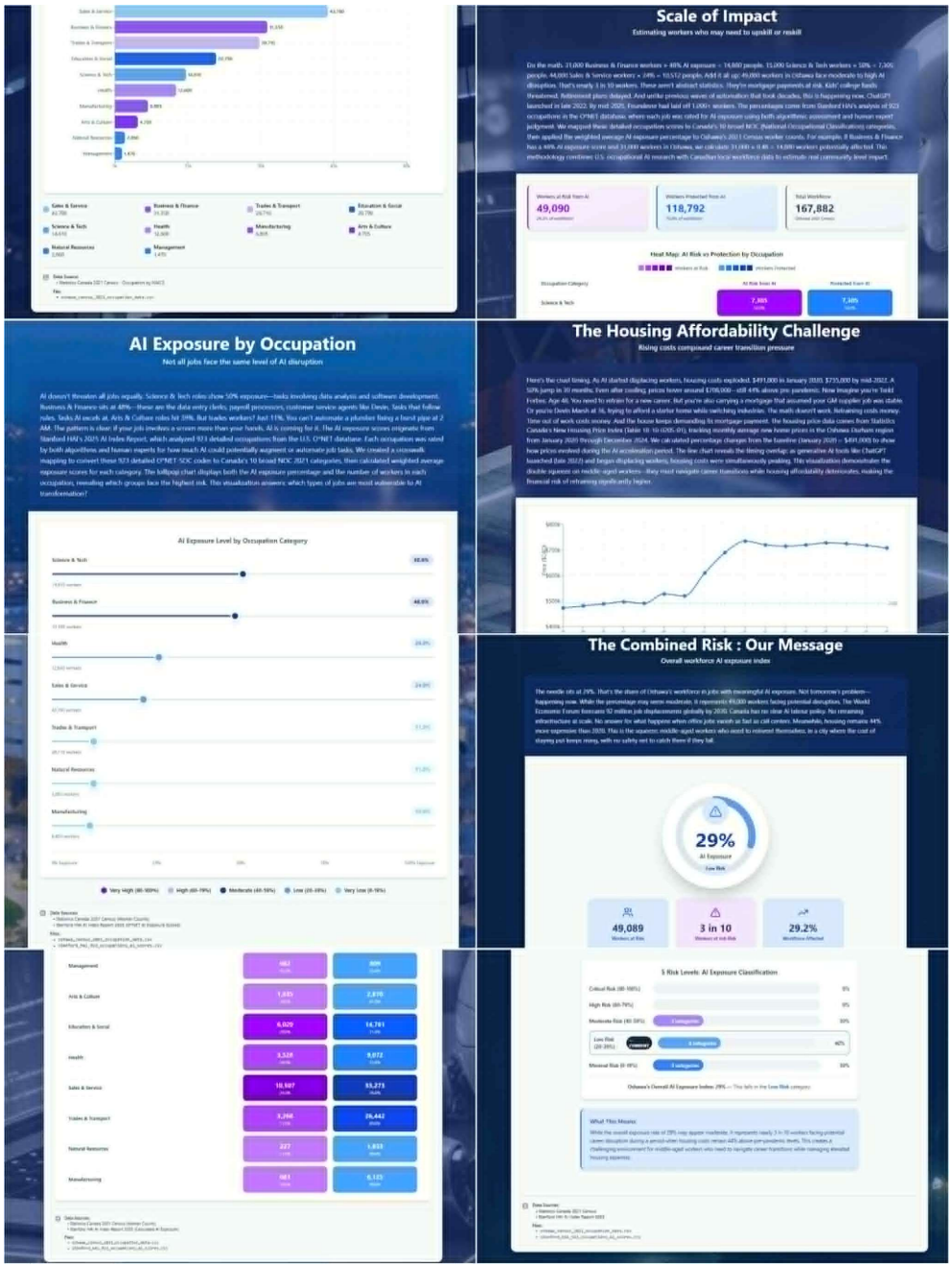

Figure 8.2: Final Output

4. Conclusion: Our Findings and Message

While the overall exposure rate of 29% may appear moderate, it represents nearly 3 in 10 workers facing potential career disruption during a period when housing costs remain 44% above pre-pandemic levels. This creates a challenging environment for middle-aged workers who need to navigate career transitions while managing elevated housing expenses.

Over the next 10-20 years, AI will continue transforming many occupations. Middle-aged workers may face increasing pressure to adapt, retrain, and develop new digital skills. If housing costs remain elevated during this transition, affordability may become even more challenging for workers dealing with career disruption. This research helps policymakers and workers better understand how AI-driven job changes and housing costs intersect, affecting financial stability and career transitions in Oshawa's workforce.

5. References:

1. Statistics Canada
2. https://www.canadianaffairs.news/2026/03/08/canadas-response-to-ai-labour-disruption-inadequate-sources-say/
3. https://www.cbc.ca/news/canada/toronto/impending-gm-oshawa-cuts-unemployment-1.7640394
4. https://github.com/openai/GPTs-are-GPTs/tree/main/data